

Medication adherence is a critical factor in managing chronic illnesses, yet many patients forget to take their prescribed doses on time. This can lead to serious health complications, reduced treatment effectiveness and increased medical costs. Traditional pillboxes rely on manual tracking, which can be ineffective with memory challenges or busy schedules.

This project aims to develop a smart pillbox using an Arduino and an MC-38 magnetic sensor to monitor whether the pillbox has been opened within a specified timeframe. If the pillbox remains unopened past the scheduled medication time, the system will trigger an alert (such as a buzzer, LED, or mobile notification) to remind the user to take their medication. By automating adherence tracking, this smart pillbox can help patients manage their medication schedules more effectively, improving health outcomes and reducing the risk of missed doses.

We proposed our solution by creating a notification system to alert stakeholder to take their medications. So we made a pill box from plexiglass with 7 slots for each day of the week, and when the lid opens, it will indicate that the person has taken their meds. If the lid isn't opened by a certain time, then the person gets the alert on their mobile device. Some features that may go beyond design requirements include our Arduino and medication box not fitting in one box.

Our feedback from experts included not only having 7 spots for days of the week but also am and pm slots since some people take medications in the morning and at night, different times of the day. The IT workers helped give us feedback on things we had not thought about previously. We struggled with connecting the wires to make it read and notify the users when the timers ends. This is something we would need to address and fix if we had time to fix the lid of the pill box. Pill boxes that are already created have 14 slots 7 for each morning of the week, and 7 for each night of the week. If we integrated are design with the LCD to tell them that they haven't taken their medication for that time, then the pre-existing pill box could be even more effective.